

Benchmark solutions for the natural convective heat transfer problem in a square cavity

J. Vierendeels¹, B. Merci¹ & E. Dick¹

¹*Department of Flow, Heat and Combustion Mechanics,
Ghent University, Belgium.*

Abstract

In this study, benchmark solutions are derived for the problem of two-dimensional laminar flow of air in a square cavity which is heated on the left, cooled on the right and insulated on the top and bottom boundaries. The temperature differences between the hot and the cold wall are large. No Boussinesq nor low-Mach number approximations of the Navier-Stokes equations are used. The ideal-gas law is used and viscosity is given by Sutherland's law. A constant local Prandtl number is assumed. An accurate and fast multigrid solver is used, which was developed in previous work. The solver is constructed so that its convergence behaviour is independent of grid size, grid aspect ratio, Mach number and Rayleigh number. Grid converged results with an accuracy of 4 up to 5 digits are obtained for a variety of Rayleigh numbers.

1 Introduction

Buoyancy-driven flows, especially in two dimensions, have been the object of thorough study for over 50 years. Most studies in the past have dealt with rectangular domains with different aspect ratios. De Vahl Davis and Jones [2] presented a study which resulted into a benchmark solution for the problem of a two-dimensional flow of a Boussinesq fluid in a square cavity which is heated on the left, cooled on the right and insulated on the top and bottom boundaries. They used the stream function-vorticity formulation of the governing equations. Chenoweth and Paolucci [1] investigated the steady-state flow in rectangular cavities with large temperature differences between vertical isothermal walls. They used the transient form of the flow equations, simplified for low-Mach number flows. Le Quére [3] studied incompressible flow in a thermally driven square cavity with a pseudo-spectral

discretization scheme based on Chebyshev polynomials. Ramaswamy and Moreno [4] computed three-dimensional buoyancy driven flows of incompressible fluids in complex geometries.

In this paper benchmark solutions are computed for a square cavity with large temperature differences between vertical isothermal walls. The Navier-Stokes equations are used without Boussinesq nor low-Mach number approximation.

The computational method gives the solution very quickly and accurately, also on very fine meshes and meshes with high grid aspect ratios. Normally the computational cost increases dramatically when finer meshes are used. Not only the computational cost for one time step increases with the number of cells but also the number of time steps needed to obtain a steady state solution increases due to the Courant-Friedrich-Lewy restriction. Furthermore, very accurate solutions need high grid aspect ratio grids in zones with steep gradients. The use of such high grid aspect ratio meshes leads to unacceptably small time steps so that often a choice has to be made between highly accurate solutions with enormous computational time and less accurate solutions with an acceptable computational time. In our method, the problem due to the grid aspect ratio is removed by the use of a line method. The low Mach number stiffness is avoided by an appropriate discretization and a local preconditioning technique. Multigrid is used as convergence accelerator. The time needed for the calculation varies linearly with the number of grid cells.

Results are shown for air, for Rayleigh numbers equal to 10^2 , 10^3 , 10^4 , 10^5 , 10^6 and 10^7 for $\varepsilon = 0.6$. Because of the large value of ε , the Boussinesq approximation is not valid.

2 Definition of the problem

We consider the flow in a differentially heated square cavity in which a temperature difference is applied to the vertical walls while the horizontal walls are thermally insulated (figure 1). This test case was the object of a previous comparison exercise for incompressible flow solvers with Boussinesq approximation [2] in which a series of reference solutions for Rayleigh numbers between 10^3 and 10^6 were produced. Here, we consider large temperature differences which impose the use of compressible solvers able to treat low Mach number flows.

For a compressible fluid, the Rayleigh number is defined as:

$$Ra = Pr \frac{g \rho_0^2 (T_h - T_c) L^3}{T_0 \mu_0^2},$$

where Pr is the Prandtl number (0.71 for air), μ the viscosity coefficient, g the gravitational constant, L the dimension of the square cavity, T_h and T_c respectively the hot and cold temperatures applied to the vertical walls, T_0 a reference temperature equal to $(T_h + T_c)/2$ and ρ_0 a reference density corresponding to T_0 . The temperature difference may be represented by a non-dimensional parameter:

$$\varepsilon = \frac{T_h - T_c}{T_h + T_c}.$$

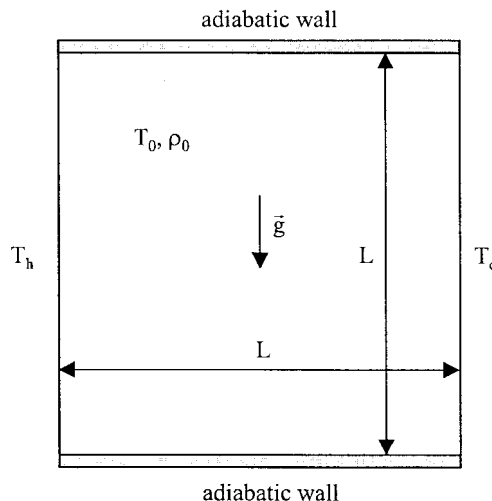


Figure 1: Geometry, initial and boundary conditions for the thermally driven cavity problem.

The heat transfer through the wall is represented by local and average Nusselt numbers Nu and $\overline{Nu}$:

$$Nu(y) = \frac{L}{k_0(T_h - T_c)} k \frac{\partial T}{\partial x} \Big|_{\text{wall}}$$

$$\overline{Nu} = \frac{1}{L} \int_{y=0}^{y=L} Nu(y) dy$$

where $k(T)$ is the heat conduction coefficient, $k_0 = k(T_0)$. In the test cases considered here, the Prandtl number is assumed to remain constant, equal to 0.71, and the viscosity coefficient is given by Sutherland's law:

$$\frac{\mu(T)}{\mu^*} = \left(\frac{T}{T^*} \right)^{\frac{3}{2}} \frac{T^* + S}{T + S}, \quad k(T) = \frac{\mu(T) C_p}{Pr},$$

with $T^* = 273$ K, $S = 110.5$ K, $\mu^* = 1.68 \cdot 10^{-5}$ kg/m/s, $C_p = \gamma R / (\gamma - 1)$, $\gamma = 1.4$ and $R = 287.0$ J/kg/K. The influence of the temperature on C_p is neglected.

The parameters defining the problem are $\mu(\mu^*, S, T^*)$, R , γ , k , T_0 , P_0 , ε , L and g . The independent dimensionless parameters appearing in the problem are: ε , γ , Pr , Ra , S/T^* , T_0/T^* and $Ma^* = v_{ref}/c_0$ with c_0 the speed of sound, $v_{ref} = Ra^{0.5} \mu(T_0) / (\rho_0 L)$ and $\rho_0 = P_0 / (R T_0)$. The Ma^* -number only affects differences in the hydrodynamic part of the pressure [1]. As long as the Mach number is very small, this parameter does not affect the flow field nor the heat transfer.

The problem is completely defined by the Rayleigh number, the value of ε , a reference state: $P_0 = 101325$ Pa, $T_0 = 600$ K, and the initial conditions:

$$\forall (x, y) \in [0, L]^2,$$

$$T(x, y) = T_0$$

$$\rho(x, y) = \rho_0$$

$$u(x, y) = v(x, y) = 0.$$

3 Computational method

3.1 Governing equations

The two-dimensional steady Navier-Stokes equations in conservative form for a compressible fluid are

$$\frac{\partial F_c}{\partial x} + \frac{\partial F_a}{\partial x} + \frac{\partial G_c}{\partial y} + \frac{\partial G_a}{\partial y} = \frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} + S,$$

where F_c and G_c are the convective fluxes, F_a and G_a are the acoustic fluxes and F_v and G_v are the viscous fluxes, in our method defined by:

$$F_c = \begin{bmatrix} 0 \\ \rho u^2 \\ \rho uv \\ 0 \end{bmatrix}, F_a = \begin{bmatrix} \rho u \\ p \\ 0 \\ \rho H u \end{bmatrix}, F_v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} + k\frac{\partial T}{\partial x} \end{bmatrix},$$

$$G_c = \begin{bmatrix} 0 \\ \rho uv \\ \rho v^2 \\ 0 \end{bmatrix}, G_a = \begin{bmatrix} \rho v \\ 0 \\ p \\ \rho H v \end{bmatrix}, G_v = \begin{bmatrix} 0 \\ \tau_{yx} \\ \tau_{yy} \\ u\tau_{yx} + v\tau_{yy} + k\frac{\partial T}{\partial y} \end{bmatrix}$$

where ρ is the density, u and v are the Cartesian components of velocity, p is the pressure, T is the temperature, H is the total enthalpy and τ_{ij} are the components of the viscous stress tensor. The source term S is given by

$$S = \begin{bmatrix} 0 \\ 0 \\ -\rho g \\ -\rho g v \end{bmatrix}$$

where g is the gravitational acceleration constant.

3.2 Solution method

An orthogonal 1024x1024 stretched grid is used with maximum grid aspect ratio equal to 80. The grid is stretched in both dimensions. The solution method is given in [5]. The solver is basically a multistage time stepping algorithm accelerated with multigrid. In order to avoid the problems coming from the acoustic terms (low speed flow), the aspect ratio and the viscous terms, these terms are treated implicitly along lines. Each pre- and postrelaxation in the multigrid cycle consist of two multistage steppings. One with lines in the x -direction and one with lines in the y -direction. The convergence behaviour of the scheme is independent of the Mach number, the Rayleigh number, ε and the grid aspect ratio [5].

4 Results

For the present study, six Rayleigh numbers, $Ra = 10^2, 10^3, 10^4, 10^5, 10^6$ and 10^7 are considered with a temperature difference parameter $\varepsilon = 0.6$. Results are computed on a 1024x1024 stretched grid, of which the maximum aspect ratio is 80. Streamline patterns and temperature distributions are shown in figures 2-3. Nusselt numbers and mean pressure values for the different Rayleigh numbers are given in table 1. The mean pressure is defined by

$$\bar{p} = \frac{1}{S} \int_S p dS$$

where S is the area of the cavity.

Ra	$\overline{Nu}$	$\bar{p}/P_0$
1E2	0.9787	0.9574
1E3	1.1077	0.9380
1E4	2.218	0.9146
1E5	4.480	0.9220
1E6	8.687	0.92449
1E7	16.241	0.92263

Table 1: Nusselt number and mean pressure for different Rayleigh numbers, $\varepsilon = 0.6$

In [5], it was shown with the use of a 384x384, 512x512, 768x748 and 1024x1024 grid that quadratic grid convergence is obtained, that very good accuracy is obtained and that the computed results are correct up to 4 or 5 digits. It was also shown that the convergence behaviour is not influenced by the number of grid cells and grid aspect ratio.

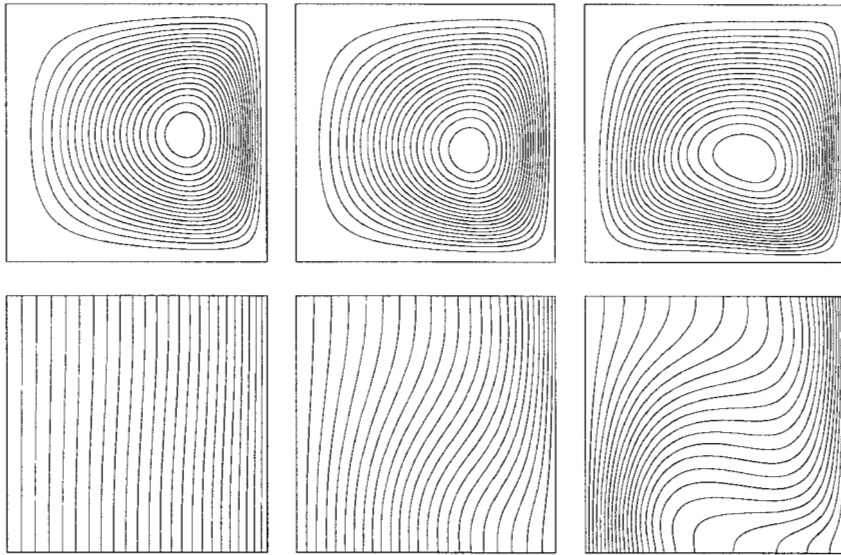


Figure 2: Streamline patterns and temperature contours, $\varepsilon=0.6$, $Ra=10^2, 10^3, 10^4$

Nusselt and velocity profiles in the midplane are shown in figures 4-5. Table 2 shows for $Ra=10^6$ and $Ra=10^7$ computed values for different dimensionless quantities: the maximum and minimum Nusselt number and the Nusselt number at $y = 0.5$ are computed at left, mid and right position, the mean pressure, maximum Mach number, minimum and maximum velocity components at the mid planes, minimum and maximum divergence of the velocity and the stream function at the center and local minima and maxima of the streamfunction.

5 Conclusion

Benchmark solutions are presented for different Ra -numbers for the square cavity problem with large temperature differences between the vertical planes with an accuracy of 4 up to 5 digits.

References

- [1] Chenoweth, D.R. & Paolucci, S., Natural convection in an enclosed vertical air layer with large horizontal temperature differences. *Journal of Fluid Mechanics*, **169**, pp. 173–2105, 1986.
- [2] De Vahl Davis, G. & Jones, I.P., Natural convection in a square cavity, a comparison exercise. *International Journal of Numerical Methods in Fluids*, **3**, pp. 249–264, 1983.
- [3] Le Quéré P., Accurate solutions to the square thermally driven cavity at high

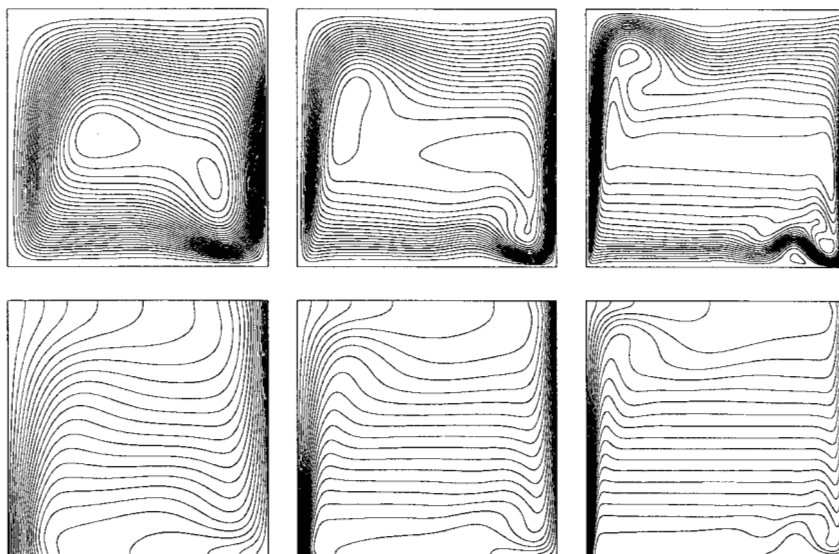


Figure 3: Streamline patterns and temperature contours, $\varepsilon=0.6$, $Ra=10^5, 10^6, 10^7$

Rayleigh number. *Computers Fluids*, **20**, pp. 29–41, 1991.

- [4] Ramaswamy, B. & Moreno, R., Numerical study of three-dimensional incompressible thermal flows in complex geometries. Part I: theory and benchmark solutions. *International Journal of Numerical Methods in Heat and Fluid Flow*, **7**, pp. 297–343, 1997.
- [5] Vierendeels, J., Merci, B., & Dick, E., Numerical study of natural convective heat transfer with large temperature differences. *International Journal of Numerical Methods for Heat & Fluid Flow* **11**, pp. 329–341, 2001.

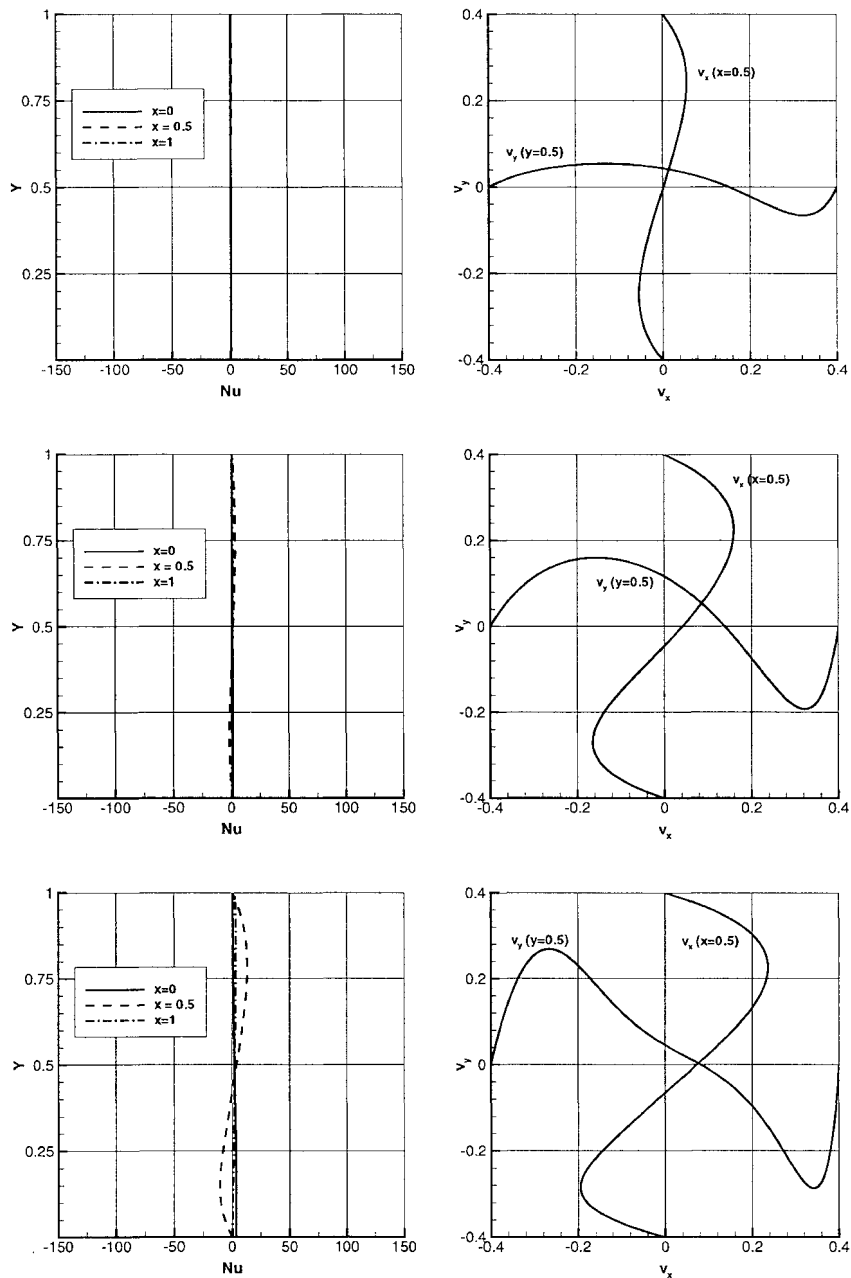


Figure 4: Nusselt profile in the vertical mid plane and velocity profiles in both mid planes, $Ra = 10^2, 10^3$ and 10^4 , $\varepsilon = 0.6$

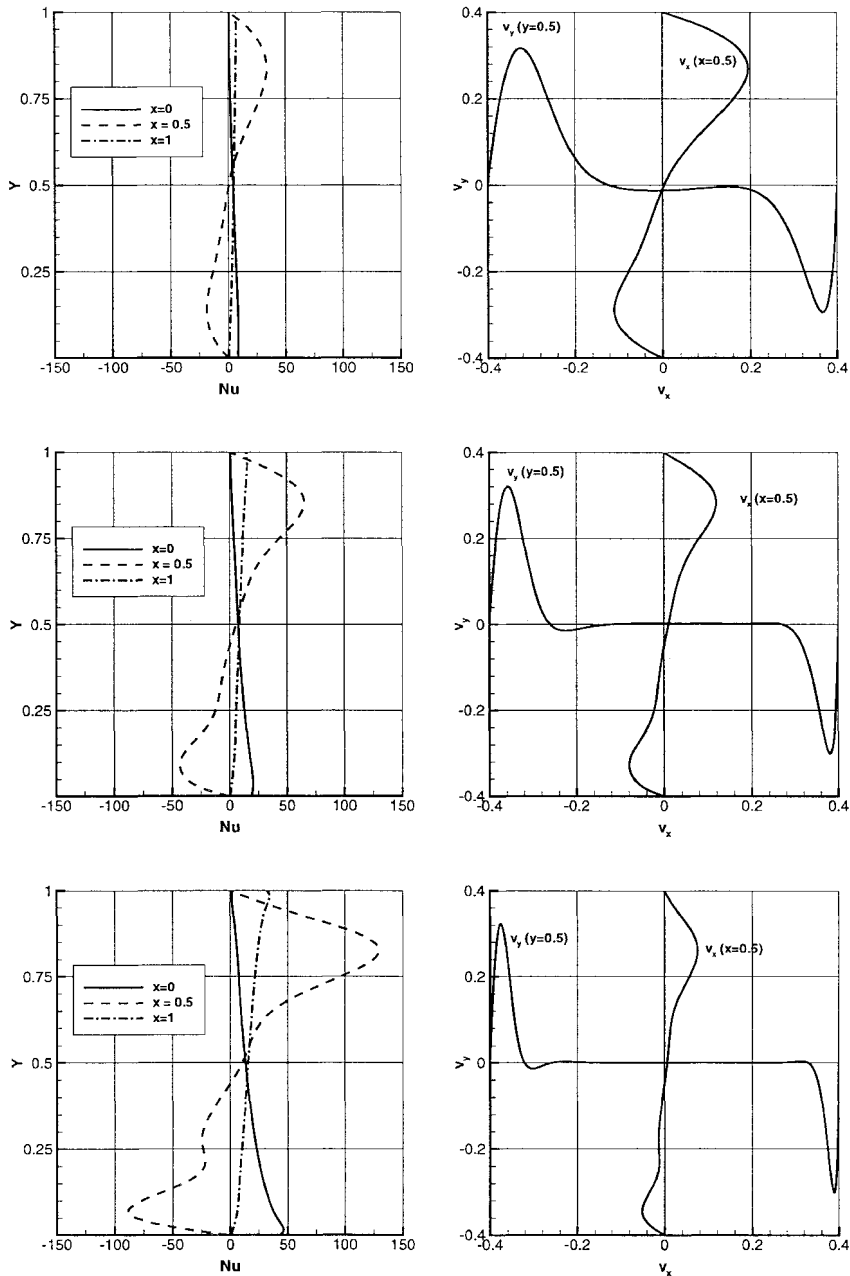


Figure 5: Nusselt profile in the vertical mid plane and velocity profiles in both mid planes, $Ra = 10^5, 10^6, 10^7, \varepsilon = 0.6$

Variable	$Ra=10^6$		$Ra=10^7$	
	Value	Position	Value	Position
Right				
Nu_{max}	15.519	(0.9676)	34.269	(0.9848)
Nu_{min}	0.758	(0)	1.089	(0)
Nu_{mid}	8.637	(0.5)	15.512	(0.5)
$\overline{Nu}$	8.687		16.240	
Left				
Nu_{max}	20.270	(0.0365)	46.379	(0.0164)
Nu_{min}	1.067	(1)	1.454	(1)
Nu_{mid}	7.459	(0.5)	13.188	(0.5)
$\overline{Nu}$	8.687		16.241	
Mid				
Nu_{max}	65.23	(0.8544)	129.35	(0.8260)
Nu_{min}	-43.56	(0.0905)	-88.43	(0.0693)
$\overline{Nu}$	8.687		16.241	
$\overline{Nu}_{conv}$	8.690		16.205	
$\overline{Nu}_{visc}$	-0.004		0.036	
$\bar{p}/P_0$	0.92449		0.92263	
Ma_{max}	3.720e-05	(0.9792, 0.3787)	1.175-04	(0.9883, 0.3754)
$v_{max} (y = 0.5)$	0.3203	(0.0537)	0.3229	(0.0305)
$v_{min} (y = 0.5)$	-0.3001	(0.9756)	-0.3011	(0.9861)
$u_{max} (x = 0.5)$	0.1193	(0.8541)	0.07490	(0.8260)
$u_{min} (x = 0.5)$	-0.07972	(0.0905)	-0.05124	(0.0693)
$(-\vec{v}/\rho \cdot \nabla \rho)_{max}$	1.0394	(0.0384, 0.0303)	1.7061	(0.0172, 0.0134)
$(\nabla \cdot \vec{v})_{max}$	1.0393	(0.0384, 0.0303)	1.7058	(0.0172, 0.0134)
$(-\vec{v}/\rho \nabla \rho)_{min}$	-0.6086	(0.9794, 0.9762)	-0.9051	(0.9903, 0.9888)
$(\nabla \cdot \vec{v})_{min}$	-0.6086	(0.9794, 0.9762)	-0.9052	(0.9903, 0.9888)
Ψ_{mid}	0.02209	(0.5, 0.5)	0.01265	(0.5, 0.5)
Ψ_{max1}	0.02351	(0.8688, 0.3926)	0.01315	(0.0989, 0.5016)
Ψ_{max2}	0.02312	(0.2081, 0.6477)	0.01211	(0.1639, 0.8211)
Ψ_{max3}	0.02126	(0.8880, 0.1458)	0.01282	(0.2798, 0.4556)
Ψ_{max4}			-0.00013	(0.8009, 0.0419)
Ψ_{max5}			0.01288	(0.8040, 0.4301)
Ψ_{max6}			0.00922	(0.8807, 0.1765)
Ψ_{max7}			0.01104	(0.9207, 0.0793)
Ψ_{max8}			0.01324	(0.9256, 0.3909)

Table 2: Computed values for $Ra = 10^6$ and 10^7 , $\varepsilon = 0.6$